

Exploring Customer Churn with Survival Analysis: Kaplan–Meier, Cox, and Accelerated Failure Time Modell

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1. Introduction

Customer churn — the event of customers discontinuing their relationship with a company — is a critical challenge in subscription-based businesses. High churn rates directly reduce revenues, increase marketing costs (as acquiring new customers is more expensive than retaining existing ones), and threaten long-term growth. For this reason, predicting not only whether but also when customers are likely to churn is of substantial practical value.

Traditional classification models (such as logistic regression or decision trees) treat churn as a binary outcome (churned vs. retained). While informative, this approach overlooks an important dimension: time until churn. Customers differ not only in their eventual status but also in how long they remain active. A method that accounts for time-to-event provides deeper insights into customer retention dynamics. This is precisely where survival analysis becomes useful. Its key advantage lies in handling censored data (customers who are still active at the end of observation) and modeling the distribution of churn over time.

The objective of this study is to apply survival analysis to a large-scale customer churn dataset from Kaggle. Specifically, we aim to:

1. Explore churn patterns over time using descriptive survival methods (e.g., Kaplan–Meier and Nelson–Aalen).
2. Compare survival across customer subgroups (e.g., gender, contract length, subscription type) using statistical tests such as the log-rank and Fleming–Harrington tests.
3. Model the effects of predictors (demographic, behavioral, and contractual factors) on churn risk using regression approaches. We consider both the semi-parametric Cox Proportional Hazards model and the parametric Accelerated Failure Time (AFT) model.
4. Evaluate model performance to determine which approach best explains churn dynamics.
5. Generate actionable business insights, highlighting high-risk customer profiles and critical time periods for retention strategies.

Through this combination of descriptive and regression-based methods, the study investigates not only *who* churns, but also *when* and *why*.

2. Exploratory Data Analysis (EDA) & Descriptive Statistics

Before conducting survival modeling, it is essential to understand the dataset and develop an initial profile of customers who are more likely to churn. Exploratory analysis not only highlights raw patterns in the data but also helps identify which variables are potentially important predictors of churn.

Dataset Description

The dataset was obtained from Kaggle (Customer Churn Dataset, <https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset>). It consists of **440,832 customers**, each described by demographic, behavioral, and contractual features.

Variables:

- **Demographics:** Age, Gender.
- **Behavioral:** Usage Frequency, Support Calls, Payment Delay, Total Spend, Last Interaction.

- **Contractual:** Tenure (time until churn or censoring), Subscription Type, Contract Length.
- **Outcome:** Churn status.

Churn & Tenure

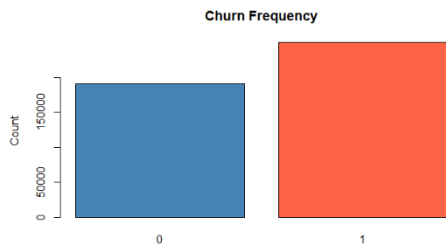


Figure1. Churn Frequency

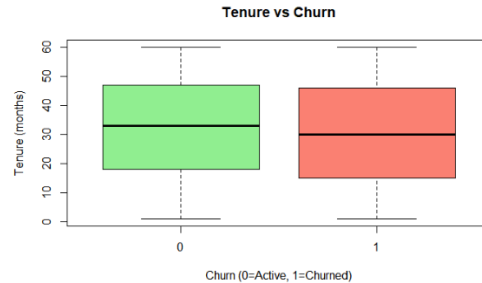


Figure2. Tenure vs Churn

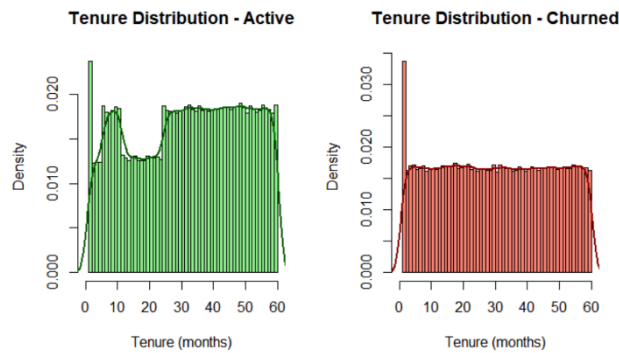


Figure 3. Distribution of Tenure - Active vs Churned

The outcome of interest is **Churn**, coded as:

- **0** = Active (censored)
- **1** = Churned (event)

The distribution of churn (*Figure 1*) shows that approximately **56% of customers churned**, while **44% remained active** at the end of observation. This high churn rate motivates the use of time-to-event models to explore not just *if* but also *when* customers leave.

Tenure measures the length of time (in months) a customer remains subscribed before churning or being censored.

- The **median tenure** is **32 months**, meaning half of the customers churn (or are censored) before ~2.5 years, and half after (*Figure2*).
- A **large spike at 1 month** indicates that many customers drop out almost immediately after subscribing, reflecting early disengagement (*Figure 3*).
- Boxplots (*Figure 2*) further show that **churned customers generally have shorter tenures** (median ~29 months) compared to active customers (~32 months), though distributions overlap substantially.

Early churn is a major business concern, as many customers leave within the first month. This validates the use of survival analysis since churn is clearly linked to timing.

Categorical & Continuous Predictors

Category	Group	Customers (n, %)	Churn Rate	Interpretation
Gender	Female	190,580 (43%)	66.7%	Women are significantly more likely to churn than men.
	Male	250,252 (57%)	49.1%	Men churn at a lower rate compared to women.
Subscription	Basic	143,026 (32.4%)	~56-58%	Subscription type shows little effect on churn.
	Premium	148,678 (33.7%)	~56-58%	
	Standard	149,128 (33.8%)	~56-58%	
Contract Length	Annual	177,198 (40.2%)	~46%	Longer contracts retain customers significantly better.
	Monthly	87,104 (19.8%)	100%	Monthly contracts result in guaranteed churn (strongest predictor).
	Quarterly	176,530 (40.0%)	~46%	Similar retention to annual contracts.

Table 1. Categorical predictors

Metric	Churners (Median)	Active (Median)	Interpretation
Age	~41	~37	Churners tend to be older.
Usage Frequency	~14	~16	Churners use the service less often.
Support Calls	~5	~1	Churners contact support more, suggesting dissatisfaction.
Payment Delay (days)	~15	~10	Churners delay payments longer.
Total Spend	~500	~750	Active customers spend more.
Last Interaction (days ago)	~15	~11	Churners interacted longer ago.

Table 2. Continuous predictors

From the analysis of both categorical and continuous predictors of churn (Table1 & Table2), we can identify clear distinctions between churn-prone and loyal customers. **Churn-prone customers** are typically older, female, and on monthly contracts; they show lower engagement through limited usage and infrequent interactions, report more dissatisfaction with frequent support calls, and display weaker financial reliability with delayed payments and lower spending. In contrast, **loyal customers** are generally younger, male, and on annual or quarterly contracts; they demonstrate higher engagement with frequent usage and recent interactions, report fewer issues with minimal support calls, and exhibit stronger financial reliability through timely payments and higher spending.

Dimension	Churn-Prone Customers	Loyal Customers
Gender	Female	Male
Contract Type	Monthly	Annual or Quarterly
Age	Older	Younger
Engagement	Less engaged (low usage, infrequent interactions)	More engaged (frequent usage, recent activity)
Satisfaction	More dissatisfied (higher support calls)	Fewer issues (lower support calls)
Financial Behavior	Delayed payments, lower spend	Timely payments, higher spend

Table 3. Profile of Churn-Prone vs. Loyal Customers4

3. Kaplan–Meier Survival Estimates and Group Comparisons

The **Kaplan–Meier (KM)** estimator was the primary tool used to explore survival probabilities of customers over time. KM provides a non-parametric estimate of the survival function, answering the question: “*What proportion of customers remain active at tenure t ?*” It is well suited here because it makes minimal assumptions about the distribution of churn times, allowing the data to “speak for itself.”

To validate and enrich these results, we also considered complementary non-parametric methods: the Nelson–Aalen estimator of cumulative hazard, the cumulative incidence function of churn, and hazard function plots. Furthermore, formal statistical tests — the log-rank and Fleming–Harrington tests — were applied to confirm whether observed subgroup differences were statistically significant. Together, these approaches ensured that findings were both visually clear and statistically supported.

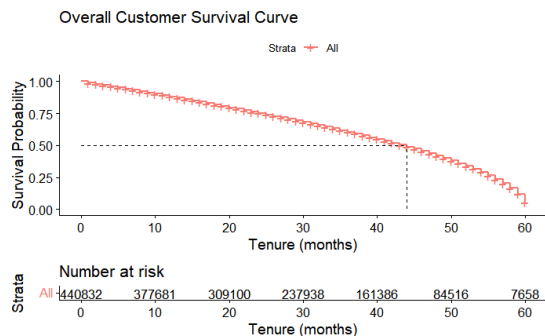


Figure 4. Overall Customer Survival Curve

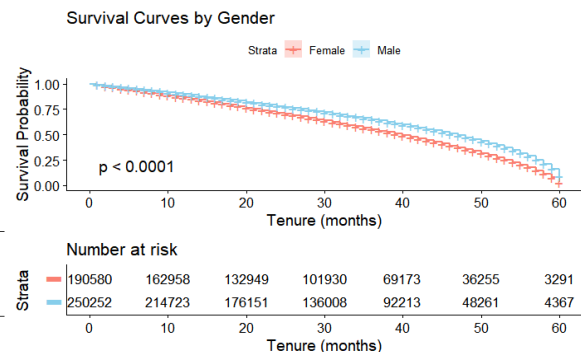


Figure 5. Customer Survival Curves by Gender

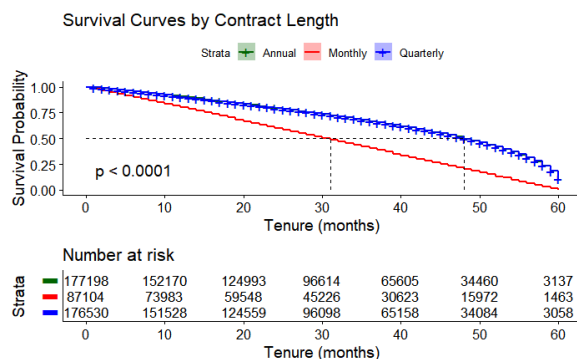


Figure 6. Customer Survival Curves by Contract Length

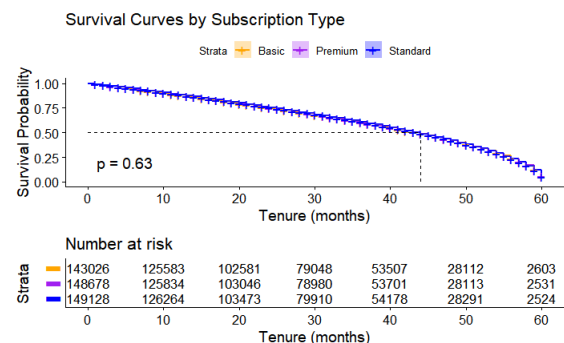


Figure 7. Customer Survival Curves by Subscription Type

Overall Survival

At tenure = 0, all customers are active ($S(0)=1$). The KM curve (Figure 4) shows a steep decline in the first month, reflecting very high early churn. By 30 months, survival probability is around 0.65, meaning roughly two-thirds of customers remain. By 60 months, survival is near zero — almost all customers churn within five years.

Supporting estimates reinforced this:

- **Cumulative incidence** indicated that ~35% of customers churned by 30 months, ~70% by 50 months, and over 95% by 60 months.
- **Nelson–Aalen cumulative hazard** showed risk accumulating gradually at first, but accelerating after ~40 months.
- **Hazard plots** revealed two critical churn phases: an early spike (month 1) and a sharp late rise (after 40 months).

Interpretation: Retention is weakest at the start of the customer journey, but churn also accelerates again after long tenures, suggesting the need for both strong onboarding and long-term loyalty strategies.

By Gender

KM curves for gender (*Figure 5*) diverged clearly: males consistently had higher survival probabilities than females. At 30 months, survival was ~70% for males versus ~63% for females. By 60 months, ~20% of males remained active compared to only ~5% of females.

Statistical tests confirmed this:

- **Log-rank test:** $\chi^2=6595$, $p<0.0001 \rightarrow$ highly significant.
- **Fleming–Harrington test ($p=1$, emphasizing early churn):** also highly significant ($p<0.0001$).

Interpretation: Female customers churn faster and more often than males. This difference is both visually and statistically strong, making gender an important predictor of churn survival.

By Contract Length

Contract length (*Figure 6*) produced the most dramatic differences. Monthly contracts showed near-total churn by 60 months, with survival dropping sharply within the first few years. In contrast, annual and quarterly contracts performed far better, both retaining ~50% of customers even at 60 months.

Tests strongly reinforced this:

- **Log-rank test:** $\chi^2=41,409$, $p<0.0001$
- **Fleming–Harrington test ($p=1$):** $\chi^2=34,900$, $p<0.0001$

Interpretation: Contract length is the strongest predictor of churn. Monthly contracts are extremely risky, while annual and quarterly contracts offer substantial retention benefits.

By Subscription Type

KM curves for Basic, Premium, and Standard subscription types (*Figure 7*) were nearly identical. All declined at the same rate: ~65% survival at 30 months and ~10–15% at 60 months.

Tests confirmed this lack of difference:

- **Log-rank test:** $\chi^2=0.9$, $p=0.63$
- **Fleming–Harrington test ($p=1$):** $\chi^2=4.3$, $p=0.10$

Interpretation: Subscription type does not meaningfully affect churn survival. Unlike contract length or gender, it offers little predictive power.

In summary, Kaplan–Meier analysis provided a clear picture of customer retention patterns: sharp early losses, steady mid-term decline, and accelerated churn in later months. Supporting non-parametric estimates (Nelson–Aalen, cumulative incidence, hazard) confirmed the timing and shape of churn risk, while statistical tests (log-rank, Fleming–Harrington) validated group differences.

Key takeaways:

- **Gender:** males retain longer, females churn earlier.
- **Contract length:** monthly contracts are by far the riskiest, annual/quarterly are protective.
- **Subscription type:** no significant effect.
- **Overall:** churn risk is front-loaded and back-loaded, requiring targeted strategies at both ends of the customer lifecycle.

4. Cox Proportional Hazards Model

Kaplan–Meier and log-rank tests show group-level survival differences but cannot adjust for multiple factors at once. For example, gender differences may partly reflect age or contract length.

Thus, the **Cox Proportional Hazards (PH) model** addresses this:

- Semi-parametric model (no assumption about baseline hazard shape).
- Estimates **hazard ratios (HRs)** for covariates.
- $HR > 1$: factor increases churn risk.
- $HR < 1$: factor decreases churn risk.
- Assumes **proportional hazards** (effects constant over time).

This makes Cox regression the workhorse of survival analysis in applied settings like churn modeling.

The Cox Proportional Hazards model (*Table 4*), used as a baseline for churn modeling, indicates that gender, contract length, age, support calls, payment delays, and engagement behaviors are significant predictors of churn. Men, higher usage, and greater spending act as protective factors, while monthly contracts, older age, frequent support calls, and payment delays increase churn risk. The model shows strong predictive performance with a C-index of 0.71, suggesting good but not perfect accuracy.

Model Fits and results:

- **Concordance (C-index):** 0.706 → model correctly orders churn times ~71% of the time.
- **Likelihood ratio, Wald, Score tests:** all highly significant ($p < 2e-16$).
- Strong predictive performance, though not perfect.

Predictor	Comparison	Hazard Ratio (HR)	p-value	Interpretation
Gender	Male vs Female	0.823	< 2e-16	Men have ~18% lower hazard of churn than women.
Contract Length	Monthly vs Annual	1.622	< 2e-16	Monthly customers face 62% higher churn risk.
	Quarterly vs Annual	≈ 1.00	0.95	No difference between quarterly and annual contracts.
Subscription Type	Premium vs Basic	≈ 1.00	NS	Not significant; no meaningful difference.
	Standard vs Basic	≈ 0.99	NS	Not significant; no meaningful difference.
Age	per year increase	1.008	< 2e-16	Each additional year increases churn hazard by ~0.8%.
Usage Frequency	per unit increase	0.999	< 0.001	Higher usage slightly reduces risk (protective).
Support Calls	per call increase	1.118	< 2e-16	Each extra support call raises risk by ~12%.
Payment Delay	per unit increase	1.019	< 2e-16	More delays increase churn hazard by ~2% per unit.
Total Spend	per unit increase	≈ 0.999	< 2e-16	Higher spending reduces churn risk slightly (protective).
Last Interaction	per unit increase	1.010	< 2e-16	More recent interaction linked to higher churn (possible dissatisfaction).

Table 4. Cox Proportional Hazard Model Results

Checking the PH Assumption

With the baseline Cox Proportional Hazards model, we also tested the proportional hazards (PH) assumption. This step is essential because the Cox model relies on the constancy of hazard ratios; if this assumption is violated, the model’s estimates and interpretations may be biased or misleading.

Using Schoenfeld residuals (cox.zph) (*Table 5*), the global test ($p < 2e-16$) indicated that the PH assumption was violated, with several covariates—including gender, contract length, support calls, and spending—showing clear deviations. This suggests that hazard ratios are not

constant over time; for example, customers on monthly contracts may face a much higher churn risk early in their tenure but less so later, while the effect of support calls may intensify as tenure increases.

Variable	Chi-Square	df	p-value	Interpretation
Gender	16.4	1	5.1e-05	PH violated (effect of gender changes over time).
Contract Length	128.0	2	<2e-16	Strong violation.
Subscription Type	12.5	2	0.0019	Violation present.
Age	28.0	1	1.2e-07	Violation present.
Usage Frequency	28.4	1	1.0e-07	Violation present.
Support Calls	251.8	1	<2e-16	Strong violation.
Payment Delay	36.1	1	1.8e-09	Violation present.
Total Spend	124.0	1	<2e-16	Strong violation.
Last Interaction	24.8	1	6.5e-07	Violation present.
GLOBAL	441.6	11	<2e-16	Overall PH assumption violated.

Table 5. Schoenfeld Residuals Test for Proportional Hazards Assumption

Stratified Cox Model

Stratification is a common remedy when the PH assumption is violated, as it relaxes the proportionality requirement for the stratified variable while still enabling valid estimation for others. To address the strong violations of the proportional hazards assumption—particularly with contract length—we applied a stratified Cox model. In this approach, the data were stratified by contract length, allowing hazard ratios for the remaining variables to be estimated within each stratum rather than assuming a constant effect across all groups.

Model Fits and results:

- Stratified by **contract length**.
- Hazard ratios for other variables estimated within each stratum.
- Concordance dropped slightly (0.706 → 0.695).
- PH test improved (global χ^2 dropped from 441 → 283), but violations persisted for support calls, spend, etc.

After applying this method, the model's concordance dropped slightly from 0.706 to 0.695, and the global PH test improved, with the chi-square statistic decreasing from 441 to 283. However, violations persisted for variables such as support calls and spending. This indicates that while stratification reduced the severity of the violations, it did not fully resolve them, suggesting that the Cox model remains informative but imperfect for capturing churn dynamics in this dataset.

5. Accelerated Failure Time (AFT) Model

The Cox PH model assumes proportional hazards, but our tests showed this assumption is violated for many predictors. In such cases, we can use the Accelerated Failure Time (AFT) model instead. The AFT model:

- Is **parametric**: it requires choosing a probability distribution (e.g., Weibull, log-normal).
- Predicts **survival time directly** rather than hazard ratios.
- Its coefficients show whether predictors **speed up or slow down** survival time.

This provides a more intuitive business interpretation: not just whether risk is higher, but **how much earlier customers churn**.

We fitted Accelerated Failure Time (AFT) models using two distributions: Weibull AFT and Lognormal AFT, each with distinct characteristics.

- In the **Weibull AFT model**, survival times are assumed to follow a Weibull distribution. This allows the hazard rate to be strictly increasing, decreasing, or constant depending on the shape parameter, making it well-suited for reliability or engineering settings where risk changes steadily over time.
- In the **Log-normal AFT model**, survival times are assumed to follow a lognormal distribution, meaning the logarithm of survival time is normally distributed. The hazard rate is non-monotonic—it rises, peaks, and then declines.

We compared the models using the **Akaike Information Criterion (AIC)**, which evaluates both model fit and complexity (*Table 6*). A lower AIC indicates a better model, and in our case, the Weibull model achieved the lowest value. Therefore, we focus on the results from the Weibull AFT model.

Model	Log-Likelihood	AIC	Interpretation
Weibull AFT	-1,161,223	2,322,473	Better fit (lower AIC, higher LogLike).
Log-normal AFT	-1,201,799	2,403,623	Worse fit compared to Weibull.

Table 6. AFT Model Comparison, Weibull AFT vs Log-normal AFT

The Weibull AFT model (*Table 7*) confirms and refines our earlier findings:

- **Strongest drivers of shorter survival:** monthly contracts, more support calls, higher payment delays, longer inactivity, older age.
- **Protective factors:** being male, higher spend, higher usage frequency.
- **Non-influential factor:** subscription type.

Variable	Coefficient	Time Ratio (exp(coef))	Interpretation
Gender (Male)	0.114	1.12	Men survive ~12% longer than women.
Contract Length (Monthly vs Annual)	-0.279	0.76	Monthly contracts shorten survival by ~24%.
Contract Length (Quarterly vs Annual)	~0.000	~1.00	No significant difference.
Subscription Type	–	–	No significant effect across types.
Age	-0.0045	0.996	Each year older reduces survival time by ~0.4%.
Usage Frequency	0.00078	1.0008	More usage slightly extends survival.
Support Calls	-0.064	0.94	Each additional call reduces survival time by ~6%.
Payment Delay	-0.011	0.99	Each extra delay shortens survival by ~1.1%.
Total Spend	0.00054	1.0005	Higher spending slightly extends survival.
Last Interaction	-0.0059	0.994	Longer since last interaction shortens survival time.

Table 7. Weibull AFT Model Results

We created three different customer profiles (*Table 8*) and simulated survival curves using the Weibull AFT model, which reveal distinct patterns in customer retention (*Figure 8*)

Profile	Gender	Contract Length	Subscription Type	Age	Usage Frequency	Support Calls	Payment Delay	Total Spend	Last Interaction
Profile1	Male	Annual	Basic	40	10	2	5	500	10
Profile2	Female	Monthly	Premium	30	20	5	2	1000	3
Profile3	Male	Quarterly	Standard	55	5	1	0	300	20

Table 8. Example Profiles Used in Weibull AFT Predictions

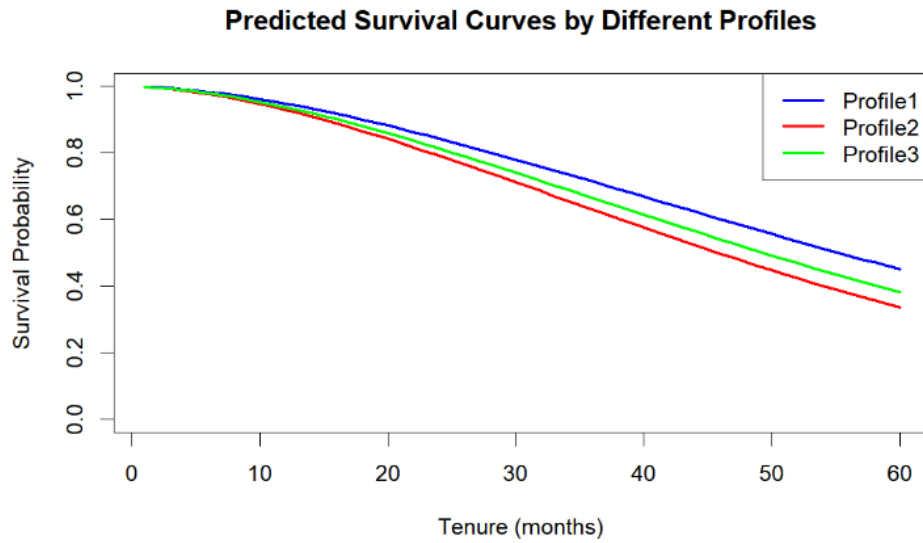


Figure 8. Predicted Survival curves by different profiles.

6. Conclusion

This study shows how survival analysis provides a powerful framework for understanding customer churn over time. Starting from non-parametric exploration and moving through hypothesis testing to Cox and AFT models, we identified key drivers of churn and gained actionable insights.

The most important findings are that **contract length, customer engagement (usage, interactions), and behavioral signals (support calls, payment delays)** are the strongest predictors of survival, while subscription type is largely irrelevant.

Methodologically, the **Weibull AFT model** emerged as the best fit, offering intuitive time-based interpretations and avoiding proportional hazards violations.

From a business perspective, the results suggest some implications:

- **Target monthly customers:** Transitioning them to quarterly or annual plans could drastically reduce churn.
- **Early engagement is critical:** The steep month-1 drop highlights the need for onboarding strategies.
- **Monitor support calls:** Frequent contact is a red flag — customer dissatisfaction must be addressed quickly.
- **Reward high spenders and engaged users:** These groups are more loyal, making them valuable for retention campaigns.
- **Tailor interventions by demographics:** Female and older customers are more at risk, suggesting customized engagement strategies.

The results emphasize the need for **early retention strategies, contract structuring, and proactive customer support** to extend customer lifetime and reduce churn.